nature portfolio

Peer Review File

Machine learning survival model for personalized prevention of catheter-related thrombosis in tumour patients

Corresponding Author: Dr Fei Ma

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
This manuscript describes the development and validation of a survival model, SM-CRT, designed to predict catheter-related thrombosis (CRT) in cancer patients. Using a large, prospective, multi-center dataset of over 30,000 patients, the authors built a model that provides both continuous risk ranking (crank) and survival distribution (distr) predictions. The model demonstrated acceptable discrimination with c-indexes around 0.7 in internal and external validation sets. The authors used the model's predictions to identify key risk factors for CRT and to stratify patients into high-risk, low-risk, and long-term risk period groups. The authors conclude that SM-CRT is a robust tool for identifying high-risk patients who may benefit from thromboprophylaxis and for guiding the optimal timing of catheter removal by identifying periods of highest risk.

Major Comments:
1. The authors mention combining different catheter types and tumor locations into collapsed factors for the model. However, the specific methodology used for this factor collapsing is not described. For methodological transparency and reproducibility, the authors should explicitly state which approach of factor collapsing was used in the study (e.g., were factors grouped based on clinical similarity, statistical analysis, or another criterion?).
2. A significant concern arises from the low incidence of the primary outcome (CRT at ~4%). With such imbalanced data, the c-index, while a standard metric for survival models, can be misleadingly high and is often insensitive to improvements in predicting the rare event. I strongly suggest the authors include a time-dependent precision-recall analysis in addition to the c-index. This analysis would evaluate the model's positive predictive value (precision) against its sensitivity (recall) over time, which is particularly valuable in imbalanced settings where correctly identifying the small number of true positive cases is paramount.
3. In the supplementary materials (Suppl Fig 3), the Brier scores for the surv.coxph model are reported as 0.405 and 0.295 in the test datasets. The manuscript text, however, refers to these as "low" Brier scores. A Brier score of 0 indicates a perfect model, while a score of 0.25 would be achieved by a non-informative model in a balanced dataset. Scores approaching 0.5 suggest very poor performance. Could the authors please clarify their interpretation of "low" in this context, as these values appear to indicate poor, rather than good, model calibration?

Minor Comments:
1/ For clarity and readability, please write out the full name for abbreviations at their first use in the main text (e.g., PICC: Peripherally Inserted Central Catheter; CVC: Central Venous Catheter; FICC: Femoral Vein Inserted Central Catheter).
2. In Figure 2a, the x-axis is not labeled with a unit of time. Please specify the unit (e.g., days, months) to allow for proper interpretation of the survival curves.
3. The legend in Figure 5 uses the term “Throm event.” Please clarify this abbreviation. If “Throm” represents “Thrombosis,” it is recommended to write its full name in the legend for clarity.

Reviewer #3

(Remarks to the Author)
The current paper builds an ensemble machine learning model SM-CRT to predict high-risk patients for thromboprophylaxis. The paper is generally well organized and written and is of good clinical interest. But there are some concerns need to address before consider for publication.
Major

1. Feature engineering techniques seems too scarce for the current data size. I only see one hot encoding and factor collapsing approaches here and I think there are a lot more potential features you can create to make it a more robust model.
2. Modelling with ensemble method. Ensemble model is known for its robustness to multi-center data, but it is also of less interpretability. The current paper I think the aim is more of proof-of-concept study to show a ML model can predict high-risk thromboprophylaxis patient after catheter intubation, instead of building the best performer for a challenge. I suggest using a simple model like xgboost or random forest to show how strong your features can predict the outcome instead of chasing the number and build a super heavy model which performs better than a simple base model but less clear with low trackability and is not scientifically elegant.
Minor
1. What does “padj” refer to in Page 2 line 44? I did not see the full term before you start to use abbreviation.
2. For Figure (a), what does the “Time” in the x-axis refer to? Is it Day after catheter as Figure 2 (b)?
3. The current paper lacks conclusion section to make it a full-length manuscript.
4. Tables reported statistical results are necessary and are absent here.
5. Standard deviation should also be reported for your statistical analysis throughout the manuscript.

Reviewer #4

(Remarks to the Author)
The paper is a (for the most part) well written work developing a survival model using ML for catheter-related prevention in cancer patients. The clinical relevance of the question is very clear and well described. The study design is well done and easy to understand. The sample size is more than sufficient for the stated objectives and the methods appear rigorous, although more detail should be provided to ensure reproducibility.

Major comments
1. The authors mention ‘multiple feature engineering methods’ and ‘factor collapsing methods’ but do not specify the techniques. This would help understand, reproduce, evaluate and improve models for readers.
2. In the methods the following is written: ‘C-index range [sic] from 0 to 1, a value of 0.5 indicates no discriminative power (…)’ (line 175). The C-index if properly defined ranges from 0.5 to 1. Pleased adapt.
3. In the section ‘kernel density estimates based on distr predictions’, it says that ‘the in-silico calculation of the distr prediction (the time distribution of the probability density function) was performed with SM-CRT’. One has to scroll back up to remember that SM-CRT is the name the authors gave to their own.. model? Pipeline? In any case, it is not clear how the method achieves this prediction.
4. The naming scheme of ‘distr’ and ‘crank’ is a bit wanting as it is not intuitively clear what these words signify, particularly ‘crank’. The latter in particularly is also not clearly specified. Is it the linear predictor (RHS of the equation) of a Cox regression model?
5. Following on point 4, it is currently unclear whether the strategy of stratification by the ‘crank’ makes sense, and particularly whether propensity score matching on the crank between crank strata is meaningful. If I understand it correctly, you fit a Cox regression model, use the continuous linear predictor to stratify individuals into groups, and then try to match individuals between these groups on the basis of how close they are in terms of the same score used for the stratification? By definition this will not work very well.
6. The comparison of hazard ratios by performing Wilcoxon tests after log-transformation is a bit unusual although not incorrect. It is however not clear in the Results section that the authors are describing log2-transformed HRs as opposed to untransformed HRs, as it is not mentioned. While the interpretation of higher log HRs correlating positively with proximity to the catheter is intuitive, it would have been good to see whether these results are robust to adjustment to patient covariates, which surely differ between the tumour sites as well.
7. Models should once be described with their full names, and not just programmer-style abbreviations, e.g. ‘surv.rfsrc’, ‘surv.coxph’ etc. The paper should be readable for a readership that does not know the particular naming conventions used by those using the software, and should be reproducible even when these particular model implementations are no longer around.
8. It is not clear what kind of features went into the survival models. Please make explicit in the Methods section.
9. Results only report on the tumour site and catheter location, but discusses catheter types and treatments in length. Without numeric results it is not possible to verify that the authors’ interpretation of the data matches the actual results.
10. The ‘model interpretation’ section should be reflected in the Methods also, not only in the Supplementary information, and the text should be improved. It should at least be mentioned what the importance ranking of features is based on. The word ‘area’ per se is insufficient to understand what is meant.
11. There is extensive attention relating to the benefits of using the ‘distr’ and ‘crank’ predictions in the Discussion’s first section, but some thing remain unclear. How can one use the PDF (of the event times?) to determine high and low risk periods at the individual patient level? The pdf, after all, is only defined in the whole sample. There is a lot of discussion on what ‘the PDF’ can do, but it is very unclear how this can concretely be done.
12. Further description on catheter characteristics is needed, starting with expanding on acronyms, to understand catheter insertion or placement, diameter and what qualifies as improper position. Is the risk associated with nutrition, treatment and tumour stage catheter dependent on type and duration, for instance?
13. Tumour stage would be an expected risk factor, as it would reflect disease progression, area, size and metastasis, given that tumour site is major risk factor. Looking at data, the difference in stages might be attributed to small sample size, competing mortality from cancer and co-morbidities in late stages, treatment type or intensity and catheter duration. However, dismissing it entirely could be a source of bias.
14. Similarly, while analysis shows that general patient characteristics such as male sex, thrombosis history, age, drinking status, and hyperlipidaemia had strong and significant hazard ratios, they are not reported nor discussed.

Minor comments
1. In the title – the word ‘personized’ is archaic and does not mean ‘personalized’, which is probably meant.
2. The overall language or sentence structure needs to be re-evaluated. Sentences tend to get long and convoluted, which can be confusing and tedious for readers.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Thank the aurhor for the detailed responses and the revisions made to the manuscript.

Major Comments:
1. While the authors state that factors were grouped based on clinical similarity and referred to Supplementary Table 2 , I was unable to find this table in the supplementary materials. Please ensure it is included. Furthermore, for the sake of transparency and reproducibility, please explicitly define any algorithm used for factor collapsing and state the specific statistical rules or thresholds (e.g., VIF or correlation coefficients) used for collinearity removal .

2. Regarding Figure 4b, for binary events at any cross-sectional time point, the concordance index is mathematically equivalent to the AUROC. The stability observed in the authors' "time-dependent c-index" during the later follow-up period suggests it is actually a cumulative c-index, which is anchored by the large number of correctly predicted pairs from the early period. To prevent potential misinterpretation, this metric should be explicitly labeled as a "cumulative c-index" throughout the text and figures. The authors should also acknowledge that this perceived stability in the late period is a result of cumulative averaging rather than a reflection of superior long-term discriminative power compared to the dynamic AUROC . (comparing to Supplementary Fig. 4)

Reviewer #3

(Remarks to the Author)
The revision addresses all my concerns

Reviewer #4

(Remarks to the Author)
I am impressed by the detail and clarity of the responses by the authors, and all my concerns have been addressed. I have no further objections to the publication of this work.

Reviewer #1:

This manuscript describes the development and validation of a survival model, SM-CRT, designed to predict catheter-related thrombosis (CRT) in cancer patients. Using a large, prospective, multi-center dataset of over 30,000 patients, the authors built a model that provides both continuous risk ranking (crank) and survival distribution (distr) predictions. The model demonstrated acceptable discrimination with c-indexes around 0.7 in internal and external validation sets. The authors used the model's predictions to identify key risk factors for CRT and to stratify patients into high-risk, low-risk, and long-term risk period groups. The authors conclude that SM-CRT is a robust tool for identifying high-risk patients who may benefit from thromboprophylaxis and for guiding the optimal timing of catheter removal by identifying periods of highest risk.

Major Comments:

1. The authors mention combining different catheter types and tumor locations into collapsed factors for the model. However, the specific methodology used for this factor collapsing is not described. For methodological transparency and reproducibility, the authors should explicitly state which approach of factor collapsing was used in the study (e.g., were factors grouped based on clinical similarity, statistical analysis, or another criterion?).

Response: Thank you for your insightful comment, which has helped improve the clarity of our manuscript. We appreciate the suggestion to explicitly describe the factor collapsing methodology used in our study.

We performed factor collapsing for tumor types and chemotherapy types. The grouping was based on clinical similarity: for tumor types, we grouped cancers with similar anatomical locations. For example, the "lung.pleura" group included lung cancer, lung sarcoma, lung neuroendocrine tumor, pleura cancer, pleura sarcoma, and pleura neuroendocrine tumor. Similarly, chemotherapy types were grouped according to their mechanisms of action. For instance, "antimetabolites" included pyrimidine analogs (such as 5-Fluorouracil, Capecitabine, and Gemcitabine) and folate analogs (such as Methotrexate).

We have provided further details on the factor collapsing approach in Supplementary Table 2 and have mentioned it in Lines 155-156 of the manuscript.

2. A significant concern arises from the low incidence of the primary outcome (CRT at ~4%). With such imbalanced data, the c-index, while a standard metric for survival models, can be misleadingly high and is often insensitive to improvements in predicting the rare event. I strongly suggest the authors include a time-dependent precision-recall analysis in addition to the c-index. This analysis would evaluate the model's positive predictive value (precision) against its sensitivity (recall) over time, which is particularly valuable in imbalanced settings where correctly identifying the small number of true positive cases is paramount.

Response: Thank you for your valuable suggestion. We fully agree that the low incidence of the primary outcome (CRT at ~4%) could result in misleadingly high c-index values. To address this, we have now incorporated a time-dependent AUPRC analysis using the tdROC package in R, alongside time-dependent AUROC for reference (see Supplementary Fig. 4 and Lines 295-302).

In Supplementary Fig. 4, we have marked the 95th percentile of patient catheter insertion time (119 days). The time-dependent AUROC and AUPRC values before this timepoint are more representative and relevant for the majority of the population. As shown in Supplementary Fig. 4, the AUROC and AUPRC values remained consistently above 0.6, and mostly above 0.7, across the three test sets, underscoring the model's performance in imbalanced settings.

However, after 119 days, both AUROC and AUPRC values become more unstable. This is because, after this timepoint, the remaining samples for AUROC and AUPRC calculations include only patients with confirmed thrombosis (event 1) or those without thrombosis at follow-up (no event 0). The number of non-thrombosis patients at follow-up (those still wearing a catheter) is very small, and a large number of censored cases are excluded from these evaluations, resulting in less stable AUROC and AUPRC values. Therefore, we have also provided the time-dependent c-index results (Figure 4b), which is a more reliable metric in this period.

3. In the supplementary materials (Suppl Fig 3), the Brier scores for the surv.coxph model are reported as 0.405 and 0.295 in the test datasets. The manuscript text, however, refers to these as "low" Brier scores. A Brier score of 0 indicates a perfect model, while a score of 0.25 would be achieved by a non-informative model in a balanced dataset. Scores approaching 0.5 suggest very poor performance. Could the authors please clarify their interpretation of "low" in this context, as these values appear to indicate poor, rather than good, model calibration?

Response: Thank you for this insightful comment, which prompted us to re-examine and clarify our interpretation of the Brier score results. We agree that, in general, Brier scores exceeding 0.25 indicate poor calibration relative to a non-informative model in a balanced setting, and our original description of these values as "low" was wrong.

Upon further inspection of the *mlr3proba* package documentation, we found that the reported Brier scores of 0.405 and 0.295 in the external test datasets correspond to *integrated Brier scores*. This metric aggregates prediction error across the entire follow-up period and is weighted by the follow-up period. As such, it reflects average predictive performance over all timepoints rather than performance within specific, clinically relevant time windows.

To better interpret these findings, we performed an additional time-dependent Brier score analysis (Supplementary Fig. 5b). We observed that the time-dependent Brier scores were consistently below 0.25 within the first 119 days (the 95th percentile of patient catheter insertion time), indicating good calibration during this period (Supplementary Fig. 5b). However, beyond 119 days, the Brier scores increased above 0.25 (Supplementary Fig. 5b), reflecting reduced predictive reliability at later timepoints. This pattern is primarily attributable to the increasing impact of censoring over time, similar to what we observed in the time-dependent AUROC and AUPRC analyses. As follow-up progresses, fewer patients remain at risk with uncensored information, resulting in greater uncertainty in prediction estimates.

In the external datasets, where the maximum follow-up durations were 255 days (External 1) and 1,271 days (External 2), the integrated Brier score is therefore disproportionately influenced by these later, less stable periods, leading to values exceeding 0.25 (Supplementary Fig. 5a).

Importantly, this does not indicate a fundamental deficiency of the model. Rather, it clarifies the domain in which the model performs most reliably: SM-CRT provides robust short-term risk prediction and stratification, whereas long-term predictions should be interpreted with greater caution due to increasing statistical uncertainty. We have revised the manuscript accordingly to accurately reflect this interpretation (Lines 302–304).

Minor Comments:
1/ For clarity and readability, please write out the full name for abbreviations at their first use in the main text (e.g., PICC: Peripherally Inserted Central Catheter; CVC: Central Venous Catheter; FICC: Femoral Vein Inserted Central Catheter).

Response: Thank you for your helpful suggestion. We have now included the full names for the abbreviations at their first use in the abstract (Lines 40-41), methods (Lines 137-139), and main text (Lines 367-369, 417-418, 648-649) for clarity and readability.

2. In Figure 2a, the x-axis is not labeled with a unit of time. Please specify the unit (e.g., days, months) to allow for proper interpretation of the survival curves.

Response: Thank you for your careful review. The unit for the x-axis is "days," and we have now added this information to Figure 2a for clarity.

3. The legend in Figure 5 uses the term "Throm event." Please clarify this abbreviation. If "Throm" represents "Thrombosis," it is recommended to write its full name in the legend for clarity.

Response: Thank you for your careful review. "Throm" refers to "Thrombosis," and we have now updated Figure 5 to directly use the full term for clarity.

Reviewer #2:

The current paper builds an ensemble machine learning model SM-CRT to predict high-risk patients for thromboprophylaxis. The paper is generally well organized and written and is of good clinical interest. But there are some concerns need to address before consider for publication.

Major

1. Feature engineering techniques seems too scarce for the current data size. I only see one hot encoding and factor collapsing approaches here and I think there are a lot more potential features you can create to make it a more robust model.

Response: Thank you for your valuable feedback. We appreciate your suggestion to expand the feature engineering techniques, and we have taken several additional steps to strengthen the model.

In addition to one-hot encoding and factor collapsing, we have incorporated normalization (scaling continuous variables to a range of 0-1), collinearity removal, and significant feature selection to improve the feature engineering process and enhance the robustness of the model (Lines 153-159, Supplementary Fig. 1). These steps standardize the feature engineering process, contributing to a more reliable model.

Regarding your suggestion to create additional potential features, we agree that linear models often require manual interaction terms or non-linear adjustments, such as cubic splines. However, we intentionally chose to use Random Forest and XGBoost models (Lines 172-177), which are non-linear models, as you recommended in your comment below. These algorithms naturally capture non-linear relationships and feature interactions without the need for manually creating additional features. Adding too many features could potentially reduce the effectiveness of the built-in feature selection and interaction capabilities of these models. Therefore, we opted to leverage the strengths of Random Forest and XGBoost rather than introducing an excessive number of new features.

2. Modelling with ensemble method. Ensemble model is known for its robustness to

multi-center data, but it is also of less interpretability. The current paper I think the aim is more of proof-of-concept study to show a ML model can predict high-risk thromboprophylaxis patient after catheter intubation, instead of building the best performer for a challenge. I suggest using a simple model like xgboost or random forest to show how strong your features can predict the outcome instead of chasing the number and build a super heavy model which performs better than a simple base model but less clear with low trackability and is not scientifically elegant.

Response: Thank you for your valuable suggestion. We agree that ensemble models, while robust, can be less interpretable and more complex. As per your advice, we have reconsidered the approach and opted for simpler models that still allow us to demonstrate the strength of our features.

We chose the following models for our task: random survival forest (surv.random.forest), XGBoost with accelerated failure time (AFT) model (surv.xgboost.aft), XGBoost with cox proportional hazards model (surv.xgboost.cox), classic cox proportional hazards model (surv.coxph), and regularized cox model with elastic net (surv.elastic.net). We then performed detailed hyperparameter tuning, shifting from broad general tuning across all models to more focused optimization for these 5 specific models (Lines 172-177).

After evaluating model performance, we found that surv.xgboost.aft demonstrated stable high performance across different test datasets. As a result, we selected this model as the final one (named SM-CRT) (Figure 4a) and updated all figures and results after Figure 4a accordingly (Figure 5-6, Supplementary Fig. 4-9, Lines 280-379).

Minor

1. What does “padj” refer to in Page 2 line 44? I did not see the full term before you start to use abbreviation.

Response: Thank you for your detailed review. "padj" refers to the adjusted p-value. We have provided a detailed method for p-value adjustment in the Methods section (Line 237-239). Additionally, we have added the full term in the abstract (Line 45) and in the corresponding results section (Line 333).

2. For Figure (a), what does the “Time” in the x-axis refer to? Is it Day after catheter as Figure 2 (b)?

Response: Thank you for your careful review. The unit on the x-axis is "days," and we have now added this information in Figure 2a for clarity.

3. The current paper lacks conclusion section to make it a full-length manuscript.

Response: Thank you for your suggestion. We have now added a dedicated conclusion section, beginning with “In conclusion,” to complete the manuscript (Lines 469–474).

4. Tables reported statistical results are necessary and are absent here.

Response: Thank you for your suggestion. We have now provided the descriptive statistical analysis of demographic characteristics for the training, prospective test, external test 1, and external test 2 datasets in Supplementary Table 1. In addition, we have included the statistical results from both univariate and multivariate Cox regression analyses in the manuscript (Figure 3 and Supplementary Fig. 3). To improve clarity and readability, we present these regression results in figure format (Figure 3 and Supplementary Fig. 3). All essential statistical information—including descriptive statistics, hazard ratios with 95% confidence intervals, and p-values—is clearly displayed within the figures.

5. Standard deviation should also be reported for your statistical analysis throughout the manuscript.

Response: Thank you for your careful review and suggestion. We appreciate the reviewer’s input. We have now consistently reported measures of variability throughout the manuscript.

For continuous variables, we have used median [IQR] as the preferred representation, as this is suitable for both normally and non-normally distributed data. Mean ± SD may not be appropriate for non-normally distributed data, so we present continuous variables as median [IQR] in Supplementary Table 1.

Regarding hazard ratios, we have also included the 95% confidence interval as a measure of variability (Lines 308-310). Additionally, we have revised the reporting of Fisher's exact test (Lines 44-47, 332-343, 355-358) and the Wilcoxon rank-sum test (Lines 271-275) to follow a more standardized format.

Reviewer #3:

The paper is a (for the most part) well written work developing a survival model using ML for catheter-related prevention in cancer patients. The clinical relevance of the question is very clear and well described. The study design is well done and easy to understand. The sample size is more than sufficient for the stated objectives and the methods appear rigorous, although more detail should be provided to ensure reproducibility.

Major comments

1. The authors mention 'multiple feature engineering methods' and 'factor collapsing methods' but do not specify the techniques. This would help understand, reproduce, evaluate and improve models for readers.

Response: Thank you for your valuable suggestion. We agree that providing more detail on the feature engineering and factor collapsing methods will improve the clarity and reproducibility of our work.

We performed factor collapsing for tumor types and chemotherapy types based on clinical similarity. For tumor types, cancers with similar anatomical locations were grouped, such as the "lung.pleura" group, which included lung cancer, lung sarcoma, lung neuroendocrine tumor, pleura cancer, pleura sarcoma, and pleura neuroendocrine tumor. For chemotherapy types, treatments with similar mechanisms were grouped, such as "antimetabolites," which included pyrimidine analogs (e.g., 5-Fluorouracil, Capecitabine, and Gemcitabine) and folate analogs (e.g., Methotrexate). We have now provided a detailed description of the factor collapsing process in Supplementary Table 2 and mentioned it in Lines 155-156 to ensure reproducibility. Additionally, we applied one-hot encoding to categorical variables and normalization to continuous variables, followed by univariable selection and collinearity removal to simplify and strengthen the model (Lines 156-159, Supplementary Fig. 1).

2. In the methods the following is written: 'C-index range [sic] from 0 to 1, a value of 0.5

indicates no discriminative power (…)' (line 175). The C-index if properly defined ranges from 0.5 to 1. Pleased adapt.

Response: Thank you for your helpful advice. We have made the necessary adjustment to the text, and the C-index is now correctly defined as ranging from 0.5 to 1 in Line 192.

3. In the section 'kernel density estimates based on distr predictions', it says that 'the in-silico calculation of the distr prediction (the time distribution of the probability density function) was performed with SM-CRT'. One has to scroll back up to remember that SM-CRT is the name the authors gave to their own.. model? Pipeline? In any case, it is not clear how the method achieves this prediction.

4. The naming scheme of 'distr' and 'crank' is a bit wanting as it is not intuitively clear what these words signify, particularly 'crank'. The latter in particularly is also not clearly specified. Is it the linear predictor (RHS of the equation) of a Cox regression model?

Response: Thank you for your thoughtful comments. As both comments relate to predictions made using the R mlr3 framework, we will address them together. We agree that the original text did not sufficiently explain the crank and distr prediction processes. We have clarified these points and made the corresponding revisions in the manuscript (Lines 178-190, 199-200, and Supplementary Methods):

(1) SM-CRT: You are correct that SM-CRT is the name of our model, which is built using the R mlr3 framework. We have revised the manuscript to clarify this in Lines 199-200. In the R mlr3 framework, when the model is used to predict a new case, it automatically generates two types of predictions: crank and distr. Further details about these predictions will be explained below and can be found in the R mlr3 documentation on Prediction Object for Survival — PredictionSurv • mlr3proba.

(2) Crank Prediction: The term crank refers to a continuous risk ranking in the mlr3 framework. It is a model-specific one-dimensional risk score produced by the underlying learner to rank individuals by risk. In the mlr3 implementation, the direction of crank is such that larger values correspond to higher risk. Its scale and sign are model-dependent. Specifically: 1) For surv.coxph, surv.glmnet, and surv.xgboost.cox, crank is the common

linear predictor. 2) For surv.xgboost.aft, crank represents the negated predicted log survival time. 3) For ranger.surv, crank is the summed cumulative hazard. We have added this explanation to Lines 178-184.

(3) Distr Prediction: The distr prediction refers to the individual-specific predicted time-to-event distribution, which is expressed through its survival function (S), cumulative distribution function (CDF), and corresponding probability density function (PDF). This prediction is automatically generated by the R mlr3 framework (PipeOpDistrCompositor — mlr_pipeops_compose_distr • mlr3proba), thus, we did not describe this process in detail in the previous manuscript. To generate the distr prediction, the process proceeds through four main stages: 1) crank prediction from a survival model, 2) calibration of the crank score within a proportional hazards formulation, 3) estimation of the baseline cumulative hazard function, and 4) analytical construction of the individual-specific survival function (distr prediction), cumulative distribution function (CDF), and probability density function (PDF).

In detail, we first use crank as the sole covariate in a Cox calibration model. For an individual with a given crank value, the hazard function is modeled as

$$h(t \mid \text{crank}) = h_0(t)\ \exp(\beta \cdot \text{crank}),$$

where $h_0(t)$ is the (unspecified) baseline hazard and β is a scalar calibration parameter. The parameter β is estimated from the training data via the standard Cox partial likelihood.

Given $\hat{\beta}$, the baseline cumulative hazard $\widehat{H_0}(t)$ is estimated nonparametrically using the Breslow estimator[1], an estimator used in semiparametric setting comparable to Kaplan-Meier estimator used in non-parametric setting[1]. Let the distinct observed event times be

$$0 < t_{(1)} < t_{(2)} < \cdots < t_{(J)}.$$

For each event time $t_{(j)}$, define the risk set and the number of events as

$$R_j = \{i: T_i \geq t_{(j)}\}, \qquad d_j = \sum_{i:T_i = t_{(j)}} \delta_i.$$

Then the Breslow increment of the baseline cumulative hazard at $t_{(j)}$ is

$$\Delta\widehat{H_0}\big(t_{(j)}\big) \;=\; \frac{d_j}{\sum_{i\in R_j}\exp\big(\hat{\beta}\cdot \mathrm{crank}_i\big)}.$$

The estimated baseline cumulative hazard is the step function

$$\widehat{H_0}(t) \;=\; \sum_{t_{(j)}\le t}\Delta\widehat{H_0}\big(t_{(j)}\big) \;=\; \sum_{t_{(j)}\le t}\frac{d_j}{\sum_{i\in R_j}\exp\big(\hat{\beta}\cdot \mathrm{crank}_i\big)}.$$

For a new individual with a given crank value, the calibrated survival function is then

$$S(\,t \mid \mathrm{crank}\,) \;=\; \exp\Big(-\widehat{H_0}(t)\,\exp\big(\hat{\beta}\cdot \mathrm{crank}\big)\Big).$$

In practice, this survival function is evaluated on a time grid $t_1,\dots,t_K$, yielding the vector

$$\mathrm{distr} \;=\; \Big(S(\,t_1 \mid \mathrm{crank}\,),\, S(\,t_2 \mid \mathrm{crank}\,),\,\dots,\, S(\,t_K \mid \mathrm{crank}\,)\Big),$$

which is the distr prediction reported by the mlr3 framework. Each component is the survival function

$$S(\,t \mid \mathrm{crank}\,) \;=\; \Pr(\,T > t \mid \mathrm{crank}\,).$$

From this, we can obtain the cumulative distribution function (CDF) from distr prediction:

$$F(\,t \mid \mathrm{crank}\,) \;=\; \Pr(\,T \le t \mid \mathrm{crank}\,) \;=\; 1 - S(\,t \mid \mathrm{crank}\,).$$

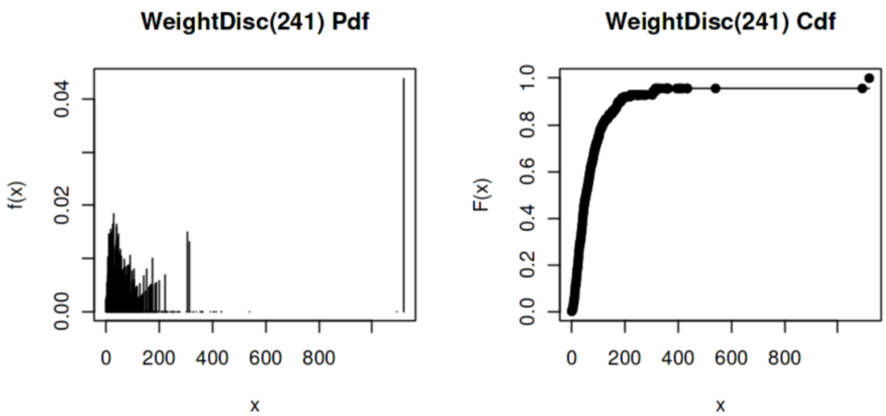
On the discrete time grid $t_1 < t_2 < \cdots < t_K$, a corresponding (discrete) probability density function (PDF) is obtained by differencing the CDF:

$$f(\,t_1 \mid \mathrm{crank}\,) \;=\; F(\,t_1 \mid \mathrm{crank}\,),$$

$$\begin{aligned} f(\,t_k \mid \mathrm{crank}\,) \;&=\; F(\,t_k \mid \mathrm{crank}\,) - F(\,t_{k-1} \mid \mathrm{crank}\,) \\ &=\; S(\,t_{k-1} \mid \mathrm{crank}\,) - S(\,t_k \mid \mathrm{crank}\,), \qquad k = 2,\dots,K. \end{aligned}$$

Thus, starting from the survival function values in distr prediction, one can derive the CDF via $F \;=\; 1 \;-\; S$, and then obtain the (discrete) PDF on the same grid by taking successive differences of the CDF.

Here is an example of CDF and PDF for an individual patient for your reference:



Since these algorithms are not original to our work, we have added a simplified explanation in Lines 184-190, and a more detailed version in the Supplementary Methods section for further clarity.

5. Following on point 4, it is currently unclear whether the strategy of stratification by the 'crank' makes sense, and particularly whether propensity score matching on the crank between crank strata is meaningful. If I understand it correctly, you fit a Cox regression model, use the continuous linear predictor to stratify individuals into groups, and then try to match individuals between these groups on the basis of how close they are in terms of the same score used for the stratification? By definition this will not work very well.

11. There is extensive attention relating to the benefits of using the 'distr' and 'crank' predictions in the Discussion's first section, but some thing remain unclear. How can one use the PDF (of the event times?) to determine high and low risk periods at the individual patient level? The pdf, after all, is only defined in the whole sample. There is a lot of discussion on what 'the PDF' can do, but it is very unclear how this can concretely be done.

Response: Thank you for your thoughtful comments. We recognize that our original description of the analysis in Figure 5 may have caused some confusion, especially regarding the use of the crank and distr predictions. We would like to address these two points together.

As discussed in our earlier response to Comments 3 and 4, the distr prediction, along with the derived cumulative distribution function (CDF) and probability density function (PDF), is patient specific. Each patient has their own distr prediction, CDF, and PDF. In Figure 5a, we presented just one example for illustration, but the analysis is based on the PDF derived from the distr prediction, which is patient specific.

Regarding the use of the 'crank' prediction for stratification in comment 4, we did not perform stratification by the 'crank' prediction (the continuous linear predictor). Instead, our focus was on stratifying by the 'distr' prediction to evaluate its potential for guiding extubation timing for patients. Specifically, we first obtain the PDF distr prediction, which gives the instantaneous probability of thrombosis throughout each catheter day. Using this, we stratify the catheter days into three periods based on the PDF: 1) Low-risk period (low PDF value), 2) High-risk period (high PDF value), and 3) Long-term period (beyond high-risk period). Patients are then stratified into three groups based on when extubation occurs: 1) Low-risk group (extubation during low-risk period), 2) High-risk group (extubation during high-risk period), and 3) Long-term group (extubation during long-term period).

This stratification allows us to compare thrombosis events per day among these three groups. As shown in Figure 5b, we found that the low-risk and long-term groups had significantly fewer CRT events per day compared to the high-risk group, based on both the training and test datasets. These results indicate that the 'distr' prediction derived from our SM-CRT model can effectively identify high-risk periods for catheter-related thrombosis (CRT) and be used to optimize catheter removal timing in clinical practice.

Regarding your concern about propensity score matching (PSM), we recognize that patient characteristics may introduce bias, as this is not a randomized controlled study. The 'crank' prediction aggregates risk across all patient characteristics, so we used PSM to balance these characteristics and reduce potential bias for the analysis of the distr prediction's ability in optimizing extubation timing.

Finally, in Figure 5c, we combine both the crank and distr predictions for more precise risk stratification. This approach not only identifies high-risk patients (based on high crank values) but also identifies high-risk periods (based on high PDF distr values). This combined approach provides more valuable guidance for clinicians in practice.

We have revised the manuscript accordingly to clarify these points (Lines 312-361).

6. The comparison of hazard ratios by performing Wilcoxon tests after log-transformation is a bit unusual although not incorrect. It is however not clear in the Results section that the authors are describing log2-transformed HRs as opposed to untransformed HRs, as it is not mentioned. While the interpretation of higher log HRs correlating positively with proximity to the catheter is intuitive, it would have been good to see whether these results are robust to adjustment to patient covariates, which surely differ between the tumour sites as well.

Response: Thank you for your careful review and constructive comments. We appreciate the opportunity to clarify both the statistical approach and the robustness of our findings.

Regarding the use of $\log_2$-transformed hazard ratios (HRs), we acknowledge that this approach may appear unusual in the context of a Wilcoxon rank-sum test. However, because the Wilcoxon test is rank-based, performing the test on either the original HRs or their $\log_2$-transformed values yields identical statistical results. In this study, the $\log_2$ transformation was applied primarily for visualization purposes.

Mathematically, hazard ratios (HRs) are derived from the Cox proportional hazards regression model, defined as $h(t \mid X) = h_0(t)\exp(\beta X)$, where $h(t \mid X)$is the hazard function given covariates X, $h_0(t)$is the baseline hazard, and βrepresents the regression coefficient. Accordingly, the hazard ratio for a one-unit change in the predictor is $\mathrm{HR} = \exp(\beta)$, meaning that effect sizes are naturally estimated on the logarithmic scale ($\beta = \log(\mathrm{HR})$), with 95% confidence intervals that are symmetric on this scale. On the original HR scale, protective effects ($0 < \mathrm{HR} < 1$) and risk effects ($\mathrm{HR} > 1$) are asymmetrically distributed, resulting in a compressed visual range for protective effects and reduced interpretability. After $\log_2$ transformation, both protective and risk effects are mapped symmetrically around zero, and the confidence intervals become more visually balanced.

Thus, the $\log_2$ transformation was used solely to improve clarity and interpretability of the figures, while all statistical inference remains unchanged.

Regarding the robustness of the observed association between higher $\log_2(HR)$ values and proximity to the catheter, we agree that adjustment for patient-level covariates is important, particularly given potential differences across tumor sites. To address this, we performed additional multivariable Cox regression analyses adjusting for all other patient covariates. These analyses demonstrated similar significant trends, supporting the robustness of our findings (Lines 275-277, Supplementary Fig. 2b–3).

7. Models should once be described with their full names, and not just programmer-style abbreviations, e.g. ‘surv.rfsrc’, ‘surv.coxph’ etc. The paper should be readable for a readership that does not know the particular naming conventions used by those using the software, and should be reproducible even when these particular model implementations are no longer around.

Response: Thank you for your valuable suggestion. We have now added the full names of each model at their first mention in the manuscript to improve clarity and readability (Lines 172–177).

8. It is not clear what kind of features went into the survival models. Please make explicit in the Methods section.

Response: Thank you for your helpful suggestion. We have now explicitly described the features included in the survival models in the Methods section (Lines 125–150). In addition, we provide a descriptive statistical analysis of these features in Supplementary Table 1, along with a detailed description of the feature engineering steps in the Methods section (Lines 153–159).

9. Results only report on the tumour site and catheter location, but discusses catheter types and treatments in length. Without numeric results it is not possible to verify that the authors’ interpretation of the data matches the actual results.

Response: Thank you for raising this important point, and we apologize for the lack of clarity in the original presentation. In addition to tumor site and catheter location, we did report numeric results for catheter types and treatments. Specifically, catheter types are included under the "catheter-related characteristics" category, and treatments are included under the "systemic treatment characteristics" category in Figure 3 and Supplementary Fig. 3, which report the corresponding univariate/multi-variate hazard ratios, 95% confidence intervals, and p-values, allowing direct verification of our interpretations. These results are also presented under the "catheter" (catheter types) and "systemic therapy" (treatments) categories in Figure 6 of model interpretation.

10. The 'model interpretation' section should be reflected in the Methods also, not only in the Supplementary information, and the text should be improved. It should at least be mentioned what the importance ranking of features is based on. The word 'area' per se is insufficient to understand what is meant.

Response: Thank you for your valuable advice. We agree that the original description of the model interpretation was insufficient and that it should be clearly described in the main Methods section. Accordingly, we have revised the manuscript and moved the model interpretation section from the Supplementary Information to the main text (Lines 208–224), with substantially improved explanations.

For model interpretation, our primary goal was to provide clinically meaningful insights into how individual features influence survival predictions. To this end, we adopted partial dependence plots (PDPs), which summarize the marginal effect of a given covariate on model predictions while averaging over the joint distribution of all other features. In the survival setting, PDPs can be naturally extended to the predicted survival function, yielding partial dependence survival curves. These curves describe how changes in a single covariate affect the estimated survival probability over time, while marginalizing over other covariates. This representation allows interpretation directly on the survival probability scale, which is intuitive and clinically relevant (Supplementary Fig. 9).

To quantify feature importance based on these survival PDPs, we defined a time-aware importance metric based on the area between partial dependence survival curves corresponding to the most favorable and least favorable values of a given feature over a prespecified time horizon (365 days) (Supplementary Fig. 9). Formally, this metric integrates the absolute difference between the two survival curves across time. This quantity has a clear interpretation: it measures the total survival probability gained or lost when a feature changes from its worst to its best value, while averaging over all other covariates.

Importantly, this area-based metric is mathematically equivalent to the difference in restricted mean survival time (RMST) implied by the two partial dependence curves. As such, it provides a time-aware, clinically interpretable measure of feature importance, rather than a purely algorithmic or scale-dependent importance score. We have clarified this explicitly in the revised Methods section (Lines 208–224).

12. Further description on catheter characteristics is needed, starting with expanding on acronyms, to understand catheter insertion or placement, diameter and what qualifies as improper position. Is the risk associated with nutrition, treatment and tumour stage catheter dependent on type and duration, for instance?

Response: Thank you for your valuable advice. We agree that additional clarification regarding catheter characteristics is important for clinical interpretation and reproducibility.

First, we have expanded all catheter-related acronyms to their full names at first mention throughout the manuscript (Lines 40–41, 137–139, 367–369, 417–418, and 648–649). In addition, we have provided detailed descriptions of catheter insertion and placement procedures, catheter diameter, and the definition of improper catheter position in the Supplementary Methods section (Catheterization method).

Regarding the potential influence of catheter-related factors on the observed associations between nutrition, treatment, and tumor stage, we acknowledge the possibility of confounding by catheter type and duration. To address this concern, we performed additional multivariable Cox regression analyses adjusting for relevant patient covariates, including catheter-related characteristics (Lines 275-277, Supplementary Fig. 3). For clarity and direct comparison, we maintained the same variable ordering between the univariate and multivariable Cox regression analyses (Figure 3, Supplementary Fig. 3). The multivariable results demonstrated trends consistent with those observed in the univariate analyses (Supplementary Fig. 3 vs. Figure 3), indicating that the associations between nutrition, treatment, and tumor stage with CRT risk are robust after adjustment for catheter-related factors.

13. Tumour stage would be an expected risk factor, as it would reflect disease progression, area, size and metastasis, given that tumour site is major risk factor. Looking at data, the difference in stages might be attributed to small sample size, competing mortality from cancer and co-morbidities in late stages, treatment type or intensity and catheter duration. However, dismissing it entirely could be a source of bias.

Response: Thank you for this thoughtful and important comment. We agree that tumor stage is a biologically and clinically plausible risk factor for CRT, as it reflects disease progression, tumor burden, metastatic status, treatment intensity, and potential competing risks.

In our analysis, tumor stage was not dismissed and was explicitly included as a risk factor. Specifically, tumor stage was examined in both the univariate Cox regression analysis (Figure 3) and the multivariable Cox regression analysis adjusting for other patient and catheter-related covariates (Supplementary Fig. 3). In addition, tumor stage was included as an input feature in the machine learning survival model. Importantly, tumor stage also emerged as a meaningful contributor in the model interpretation analyses (Figure 6 and Supplementary Fig. 9).

14. Similarly, while analysis shows that general patient characteristics such as male sex, thrombosis history, age, drinking status, and hyperlipidaemia had strong and significant hazard ratios, they are not reported nor discussed.

Response: Thank you for your helpful suggestion. We have now mentioned the results for general patient characteristics in the Results section (Lines 266–268) and expanded the corresponding discussion to interpret these findings in greater detail in the Discussion section (Lines 408–415).

Minor comments

1. In the title – the word ‘personized’ is archaic and does not mean ‘personalized’, which is probably meant.

Response: Thank you for your careful review. We have corrected the wording in the title to use “personalized.”

2. The overall language or sentence structure needs to be re-evaluated. Sentences tend to get long and convoluted, which can be confusing and tedious for readers.

Response: Thank you for your advice. We have carefully revised the manuscript to improve language clarity, simplify sentence structure, and enhance overall readability for readers.

References

1 Xia, F., Ning, J. & Huang, X. Empirical Comparison of the Breslow Estimator and the Kalbfleisch Prentice Estimator for Survival Functions. *J Biom Biostat* **9** (2018). https://doi.org:10.4172/2155-6180.1000392

Reviewer #1:

Thank the aurhor for the detailed responses and the revisions made to the manuscript.

Major Comments:

1. While the authors state that factors were grouped based on clinical similarity and referred to Supplementary Table 2 , I was unable to find this table in the supplementary materials. Please ensure it is included. Furthermore, for the sake of transparency and reproducibility, please explicitly define any algorithm used for factor collapsing and state the specific statistical rules or thresholds (e.g., VIF or correlation coefficients) used for collinearity removal .

Response: Thank you for your careful review and valuable feedback. We sincerely apologize for the omission of Supplementary Table 2 in the originally submitted supplementary materials. The table has now been included in the "Supplemental information" file under the Supplementary Tables section.

Regarding the factor collapsing procedure, collapsed binary variables were constructed such that the collapsed variable was coded as 1 if any of the corresponding component variables equaled 1, and 0 otherwise. This approach allowed us to reduce dimensionality while retaining clinically relevant information from the original variables.

For univariable selection and collinearity removal, candidate predictors were first evaluated using univariable Cox proportional hazards regression. Variables with statistically significant associations ($p < 0.05$) were retained for further consideration. Pairwise Pearson correlation coefficients were then calculated among the retained variables. When two variables demonstrated high correlation ($|r| > 0.80$), the variable with the smaller absolute log-hazard ratio in the univariable Cox analysis was removed to reduce redundancy while preserving stronger prognostic contribution.

These procedures have been clarified in the revised manuscript (Lines 163–173) to enhance transparency and reproducibility.

2. Regarding Figure 4b, for binary events at any cross-sectional time point, the concordance index is mathematically equivalent to the AUROC. The stability observed in the authors' "time-dependent c-index" during the later follow-up period suggests it is actually a cumulative c-index, which is anchored by the large number of correctly predicted pairs from the early period. To prevent potential misinterpretation, this metric should be explicitly labeled as a "cumulative c-index" throughout the text and figures. The authors should also acknowledge that this perceived stability in the late period is a result of cumulative averaging rather than a reflection of superior long-term discriminative power compared to the dynamic AUROC . (comparing to Supplementary Fig. 4)

Response: Thank you for this insightful and technically important comment. We agree that the reported metric in Figure 4b was calculated in a cumulative manner. Specifically, comparable subject pairs up to each time point were included in the calculation, which corresponds to a cumulative C-index rather than a dynamic (incident) time-specific C-index.

To avoid potential misinterpretation, we have revised the manuscript and figures accordingly. The metric is now explicitly labeled as “cumulative C-index” in the Lines 338-344, Figure 4b, and the Figure 4b legend. In addition, we have clarified in the Results section that the perceived stability observed during later follow-up reflects cumulative averaging anchored by earlier correctly predicted pairs and does not indicate superior long-term discriminative performance (Lines 338-353).

Reviewer #3:

The revision addresses all my concerns

Response: Thank you for your positive feedback.

Reviewer #4:

I am impressed by the detail and clarity of the responses by the authors, and all my concerns have been addressed. I have no further objections to the publication of this work.

Response: Thank you for your acknowledgment and supportive feedback.